

Requirement Statement	Corresponding Entity or Relationship	Implementation Elements (Attributes & Keys)	Cardinality & Participation	Brief Justification
PRIMARY REQUIREMENTS				
"The clinic needs to manage doctors, patient records, appointments, and services."	Doctor, Patient, Appointment, Medical_Service	PKs: employee_id, health_id, appointment_id, service_code.		These represent the core objects of the system. Each requires a unique identifier (Primary Key) to ensure data integrity and avoid duplicate records.
"Each doctor can treat multiple patients and provide multiple services."	"The system should track appointments, including the involved doctor, patient, and services performed."	Appointment table is a weak entity and we used it as an associative entity as well to link doctors and patients FKs: doctor_employee_id, patient_health_id (in Appointment).	Doctor and patient have a many to many relationship so we used as associative entity for that 1:M (Doctor/Pt to Appt) Participation in appointment is not mandatory	Doctors and patients are linked through the Appointment. A single doctor or patient can participate in many appointments over time, while an appointment belongs to exactly one of each.
"The system should track appointments, including the involved doctor, patient, and services performed."	Associative/ Junction Entity: Appointment_Service	Elements: appointment_id (FK), service_code (FK).	1:M (Appt to Junction) Mandatory	Since an appointment must link to " one or more " services, this junction table is required. It allows multiple service rows to be associated with a single appointment record.
"The clinic also manages medical devices that can be reserved and assigned to appointments.","Devices can be reserved for specific appointments." ,"Equipment reservations must be tied to appointments where relevant."	Junction Entity: Equipment_Reservation	Elements: appointment_id (FK), serial_number (FK), reservation_time.	M:N (Appt to Equipment) Optional	This entity tracks the specific assets assigned to a time slot. It is optional because not all clinic visits require specialized medical devices reserved.

"Services may have required equipment/devices associated with them."	Junction Entity: Service_Equipment	Elements: service_code (FK), serial_number (FK).	M:N (Service to Equipment) Optional	The associative entity suggests what equipment is needed for a specific service. It is optional to allow for services (like consultations) that may not have any particular device associated with them.
DETAILED REQUIREMENTS				
Doctor: "Unique ID, Name, Specialization, and Contact Info."	Entity: Doctor	employee_id (PK), full_name, specialization, contact_info.	M:N (to patients through appointments) Optional	Doctors and patients are linked through the Appointment. A single doctor can participate in many appointments over time, while an appointment belongs to exactly one doctor. Participation is optional because Doctors can be added in the system without any appointments, mostly in the case of new doctors.
Patient: "Unique ID, Name, DOB, and Contact Info."	Entity: Patient	health_id (PK), name, date_of_birth, contact_info.	M:N (to doctors through appointments) Optional	Patients and doctors are linked through the Appointment. A single patient can book many appointments over time, while an appointment belongs to exactly one patient. Participation is optional because patients can be registered in the system without any appointment booking.
Medical Service: "Unique code, Name, Description."	Junction Entity: Medical_Service	Elements: service_code (PK), service_name, description	M:N (Service to Equipment) Optional	Many medical services will require different equipment over time and vice versa. We believe there can be medical services without any need for equipment e.g. In the case of a consultation. Hence, Optional participation.
Equipment: Serial number, Type, Status, Model.	Entity: Equipment	serial_number (PK), type, availability_status, model.	M:N (Equipment to Medical service) Optional,	Many equipments will be require for different medical service over time. We believe there can be medical services without any need for equipment e.g. In the case of a consultation. Hence, Optional participation.

Appointment: Appointment_id, Status, Date/Time, Notes, Visit Reason.	Entity: Appointment	appointment_id (PK), data_time, status, visit_reason, notes, doctor_employee_id (FK), patient_health_id (FK)	M:1 (Appt to Dr/Pt) Mandatory	For there to be an appointment, there must be a doctor or patient involved. So participation is mandatory.
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